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(54) **SEPARATION OF METAL IONS BY LIQUID-LIQUID EXTRACTION**

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(57) **ABSTRACT**

Provided herein are separation processes for metal ions present in aqueous solutions based on methods involving liquid-liquid extraction. The separation process involves a chelator that can selectively bind to at least one of the metals at a relatively low pH. This can be used, for example, for recovery and purification of actinides from lanthanides, separation of metal ions based on their valence, and separation of metal ions based on the pH of the extraction conditions.